Migration Networks as a Shock Transmission Infrastructure

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Abstract

Migration networks are not just conduits for labor mobility but also act as infrastructure through which economic shocks can propagate across borders. This paper studies the Mexico-United States migration network as a bipartite shock transmission system, using the Illinois Budget Impasse (July 2015-July 2017) as a natural experiment. We construct a municipality x state migration weight matrix from Matrícula Consular (MCAS) data and apply betweenness and eigenvector centrality measures to identify structural bridge municipalities, through which cross-state shock transmission is most likely to occur. A state co-exposure matrix ($M^\top M$) is derived to identify which US states share sending communities with Illinois and should therefore absorb rerouted migration flows. Using a synthetic control method, we find that Illinois remittance outflows fell by approximately $400M relative to their counterfactual following the onset of the impasse, while a before-after exposure test shows that municipalities more reliant on Illinois routing experienced remittance growth roughly 6 percentage points below the national trend during 2015-2017. Furthermore, within treated municipalities, migration shifted toward states with higher $M^\top M$ co-exposure with Illinois, suggesting that a localized fiscal shock in one US state propagates through shared migrant-sending communities, affecting remittance flows in Mexican municipalities and redirecting migration toward structurally co-exposed states even in the absence of any direct policy linkage between them.

1 Introduction

Migration is not simply a bilateral relationship between a sending community and a receiving country; it is a network phenomenon. People who migrate usually maintain their ties with their origin communities, which makes those movements not only an improvement in migrants' welfare but also a direct source of welfare transmission from abroad to the origin country. The aforementioned well-being gain usually comes from direct income transfers via family members living in the destination country back to those who stayed at origin community. These additional transfers typically represent a significant share of total household income at origin, making remittances a highly important component of household resources.

According to The World Bank [World Bank, 2024], Mexico was, in 2024, the second largest remittance recipient country by total value of transfers in US dollars, just after India. This characteristic makes the country and its families highly dependent on such earnings, which gives relevance to the research question this paper is exploring: under a localized shock in the US, how are remittances towards Mexico affected, and how is the shock propagating through the migration network?

Around 10% of Mexicans live outside of their home country and, among those, in 2024, 96% were located in the United States [Instituto de los Mexicanos en el Exterior, 2024]. This massive share of people between the two countries creates an innumerable number of links between Mexican municipalities and US states, creating a perfect setting for this study.

Community networks affect migration decision allocations [McKenzie and Rapoport, 2010]; therefore, Mexicans moving to the US are neither going to allocate themselves in the exact same city, nor will they randomly allocate themselves all over the American surface. Following this logic, migrants from a Guerrero village, according to standard international migration literature, will receive higher amenity utility if they migrate closer to other former Guerrero residents, either for family ties, cultural proximity, etc.; indeed, the allocation patterns partially follow this logic. Building on this, the same Mexican municipality often feeds multiple US states simultaneously, and each US state draws from an overlapping set of Mexican origin communities. This overlapping structure means that a shock materialized as a recession, a policy change or a fiscal crisis, confined to one US state does not stay there. It travels backward through the network to the Mexican municipalities exposed to that state, and then forward along whatever other corridors those municipalities maintain, eventually arriving in US states that were never directly touched by the original shock.

This paper studies that transmission mechanism using the Illinois Budget Impasse (July 1, 2015 - July 10, 2017) as a natural experiment. The impasse left Illinois without an operating budget for 736 days, the longest such gap in US history. It disrupted state-funded construction, halted contractor payments, and froze social services, hitting sectors where Mexican immigrants are concentrated. [Illinois Office of the Comptroller, 2017]

The impasse offers three identification advantages:

- **exogenous** to migrant decisions: a political stalemate over pension reform and tax policy, not immigration policy. Mexican migrants played no causal role;
- **geographically isolated**: no other large-corridor state experienced an equivalent fiscal shock; and
- **precisely timed**: exact start (July 1, 2015) and end (July 10, 2017) dates create a sharp pre/post window in annual MCAS data.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature, Section 3 describes the data and outlines the methodological framework, including the construction of the migration weight matrix and exposure variables. Section 4 builds the bipartite migration network and derives betweenness and eigenvector centrality measures, while also applying the Louvain community detection algorithm to the state co-exposure matrix. Section 5 introduces the Illinois Budget Impasse as the shock of interest and discusses

its exogeneity, geographic specificity, and precise timing. Section 6 presents the empirical tests, and Section 7 and 8 discuss the results and conclude with policy implications and limitations.

2 Literature Review

The main relevant paper for this research is [Caballero et al., 2023], who study the Great Recession as a large and geographically differentiated labor demand shock in the US. The paper traces the effects of the shocks on Mexican municipalities with pre-existing network ties to the hardest-hit US regions. Using the same data as this article does, they find that return migration rose, emigration fell, and remittance flows declined for municipalities highly connected to the shock locations.

Our paper extends this work in two main directions. Caballero et al. focus on the direct effects of the labor demand shock on the connected Mexican municipalities but, due to the context and nature of their labor impact, they did not explore the potential effects on other states within the US. This paper fills the propagation gap by exploring to what extent a specific localized shock affects other non-related states through the Mexican migration networks, given its bipartite structure.

Related to the previous extension, our paper addresses a single localized shock that affected only one state, Illinois, instead of the whole US. This setting allows us to understand the impact through the network, as we assume other US states were not affected by it.

The use of remittances as an economic transmission mechanism is not new. A substantial literature studies remittances as a device to smooth consumption in origin households ([Yang and Choi, 2007] and [Amuedo-Dorantes and Pozo, 2011]).

The use of graph-theoretic centrality measures to construct treatment variables or identify causal pathways is relatively new in economics. Some previous research using network centrality includes [Acemoglu et al., 2012] and [Jackson et al., 2012]. Even more aligned with our study topic, [Blumenstock et al., 2019] use betweenness centrality in mobile money networks to identify the transmission hubs for economic shocks. Adapting this research to our setting, municipalities with high betweenness centrality are structurally positioned between US states and are therefore the most likely propagation nodes for the shock.

Our contribution is to use the network's own structure, bipartite centrality, community membership, and the $M^T M$ state co-exposure matrix, to predict where the shock should propagate before we look at whether it did. This turns a purely empirical question into a disciplined test: if the network predicts propagation from Illinois to California through bridge municipalities, and if we then find an effect for California in exactly those municipalities with the highest predicted exposure, the causal chain is credible.

3 Data and Methodology

This paper uses annual Matricula Consular (MCAS) data from 2010-2024, which records the number of Matricula Consular IDs issued per (US state, Mexican municipality) pair each year. These consular IDs serve as a proxy for registered migrant presence. We complement

this with DS2 quarterly remittance outflow data per US state and DS1 actual remittance inflow data at the Mexican municipality level from Banxico. The migration weighting matrix W is a municipality $\times$ state matrix where entry W_{ij} is municipality i's average share of US state j's total Matricula IDs, grand-total normalised across all years (excluding COVID years 2020-2021) and averaged over the panel.

The methodological framework combines network analysis with a before-after empirical strategy. The weight matrix defines a bipartite graph linking Mexican municipalities to US states, on which we compute betweenness and eigenvector centrality to characterize each municipality's structural role. We also derive the state co-exposure matrix $M^\top M$, where each entry captures the similarity between states' sending-community profiles, and apply the Louvain algorithm to identify groups of states exposed to similar shocks. The empirical analysis consists of three tests: a synthetic control validation of the Illinois shock, a before-after exposure regression for municipality-level remittance growth, and a cascade ordering test for within-municipality migration reallocation. Each test evaluates a different step in the proposed transmission mechanism by comparing outcomes during 2015-2017 with the pre-period baseline (2012-2014).

4 Network Construction

4.1 The Bipartite Migration Graph

The migration network is naturally bipartite, meaning that nodes split into two disjoint sets (Mexican municipalities $\mathcal{M}$ and US states $\mathcal{S}$), and edges only run between sets, never within them. An edge (i, j) with weight W_{ij} exists whenever municipality i sends a positive share of migrants to state j. One assumption underlying the bipartite structure is that within-Mexico migration is not modelled; no within-country income transfers are assumed to offset a US shock.

Formally, let $W \in \mathbb{R}^{|\mathcal{M}|\times|\mathcal{S}|}$ be the grand-total-normalised average migration weight matrix. The bipartite graph $G = (\mathcal{M} \cup \mathcal{S},\ E,\ W)$ has an edge for each non-zero entry of W, with that entry as the edge weight.

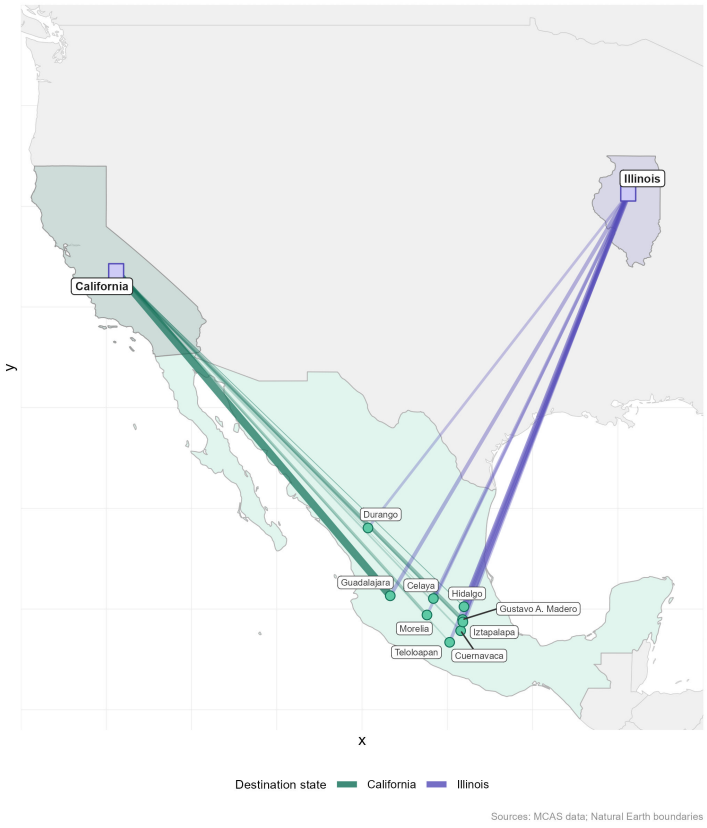


Figure 1: Migration network: top 15 MX municipalities' links to IL and CA

Figure 1 represents a simplified version of only two US states' links to their major exposed Mexican municipalities. The figure represents the paper's idea visually. If the Illinois shock propagates along the network, would it affect the state of New York, even if the latter was not directly linked to the affected origin municipalities?

4.2 Centrality analysis

Centrality measures indicate which municipalities are structurally important in the network, not simply because they send many migrants to a given state, but because of their position within the whole graph. Two measures are especially relevant here.

Betweenness centrality counts how often a node lies on the shortest path between any other pair of nodes. In our bipartite graph, all paths between US states must pass through at least one municipality, so a municipality with high betweenness is one that serves as an unavoidable intermediary across many state pairs. These are the bridge municipalities through which an Illinois shock can propagate toward California: removing them from the network would disconnect otherwise linked migration corridors.

Eigenvector centrality scores a node by the centrality of its neighbours. A municipality that sends migrants to several large, well-connected states scores highly even if its raw weight to any single state is modest. We use the standardised eigenvector score as a

continuous measure of network embeddedness, capturing a municipality's exposure to the broader migration system rather than just its bilateral tie to Illinois.

Together, the two measures serve distinct roles: betweenness identifies the municipalities most likely to transmit the shock across state boundaries, while eigenvector centrality measures how deeply embedded a municipality is in the network core, and therefore how large the downstream consequences of any shock hitting it are likely to be.

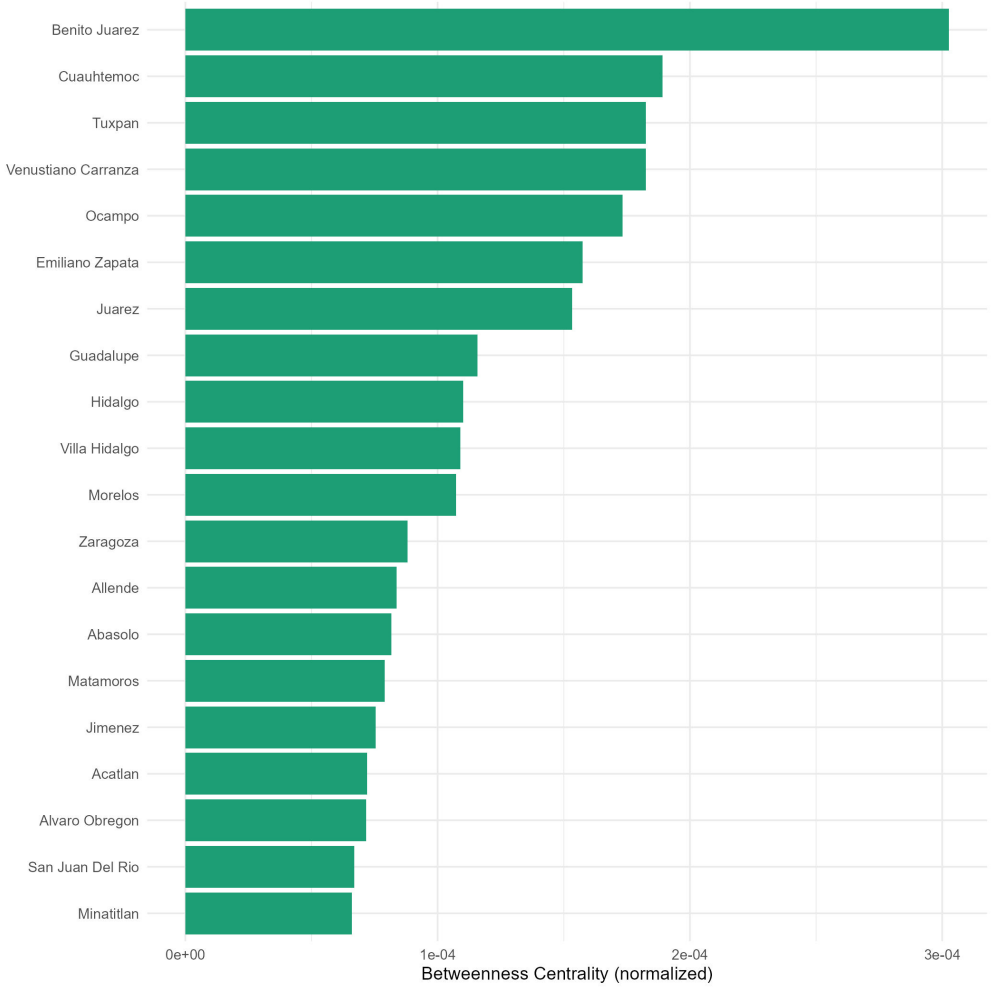


Figure 2: Top 20 MX Municipalities by Betweenness Centrality

The above Figure 2 depicts a sharply right-skewed distribution. Benito Juárez village obtains almost double betweenness score then the second listed municipality. This measure implicitly tells that Benito Juárez is precisely an 'unavoidable intermediary'. Shocks affecting that village are likely to have a direct propagation towards other US states. Moving to the following communities, Cuauhtémoc, Tuxpan, Venustiano Carranza, Ocampo, and Emiliano Zapata collectively represent a relay layer: if a shock propagates beyond Benito Juárez, these nodes are the most likely onward conduits between US states. Figure 2 also informs about the fact that structural brokerage capacity falls off quickly after the top few nodes, meaning the concentration of betweenness in very few municipalities.

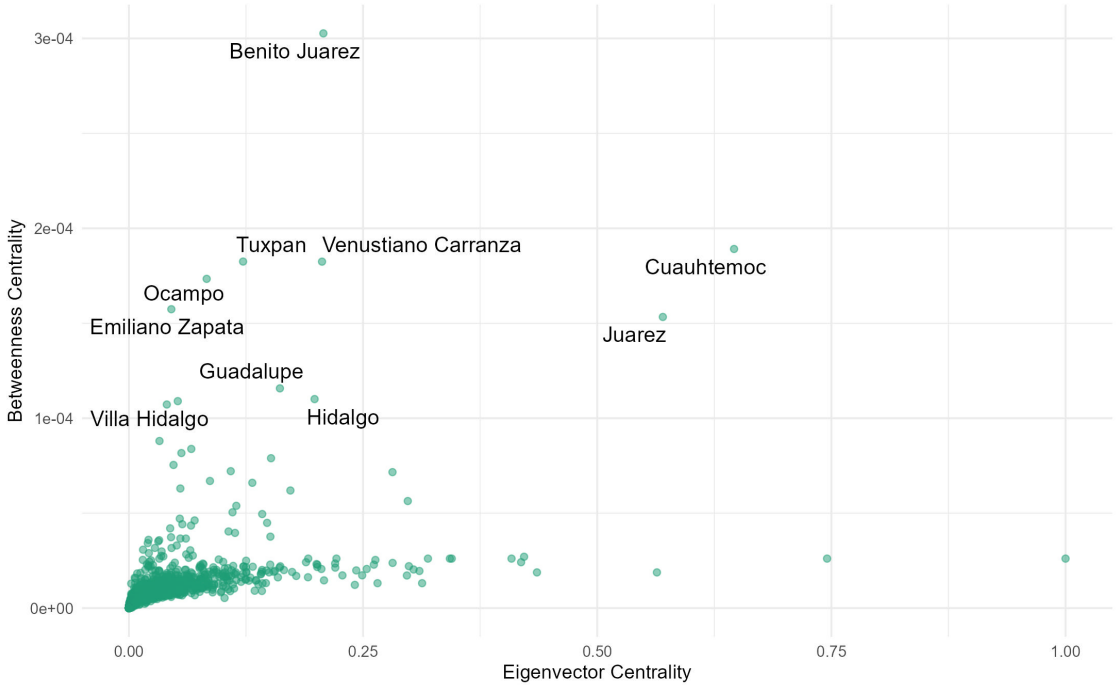


Figure 3: Centrality in the MX-US Migration Network

Figure 3 jointly plots the betweenness and eigenvector centrality across all municipalities in the network. The two measures are largely orthogonal, empirically validating the theoretical distinction made between them.

We can interpret the graph into four different sections:

- High betweenness, low-to-moderate eigenvector: Benito Juárez is the clearest example, its betweenness is extremely high (as seen in Figure 2), yet its eigenvector score sits below 0.25. This means that it functions as a structural bridge across many US state pairs, but it does not send migrants primarily to the largest, most connected destination states.

- High eigenvector, moderate betweenness: Cuauhtémoc and Juárez stand out in this region. Both score above 0.75 on eigenvector centrality, meaning they send immigrants to large and well connected states, yet their betweenness score, although also high, is far below Benito Juárez. These municipalities are deeply embedded in the network and would amplify any shock reaching them but are less likely to be the path through which a shock first crosses state boundaries.

- Mass at origin: The vast majority of the municipalities cluster around (0,0), confirming that the importance of the network is highly concentrated in a small elite of sending communities. The empirical finding helps us validate the identification strategy in our setting as there is a clearly separable treatment and control groups.

- Outlier with eigenvector almost equal to one: This region is almost entirely specialized in sending migrants to the most connected US states, yet never serving as a relay between state pairs.

4.3 Community Detection on the $M^\top M$ State Co-Exposure Matrix

When two US states draw heavily from the same set of Mexican municipalities, a shock in one state will be transmitted to the other via those shared sending communities. The inner product $M^\top M$ measures this overlap directly:

$$[M^\top M]_{jk} = \sum_{i \in \mathcal{M}} W_{ij} \cdot W_{ik}$$

A large (j, k) entry means that states j and k recruit from many of the same Mexican municipalities. In practice, we normalize each state's weight vector to unit length before computing the inner product, so the scores measure how similar the two states' sending-community profiles are rather than just how large their absolute flows are. California's overlap with Illinois reflects shared structure, not the fact that California is enormous. We then treat $M^\top M$ as a weighted adjacency matrix between US states and run Louvain community detection [Blondel et al., 2008] on it. Communities are groups of states that share sending municipalities and are exposed to each other's shocks as a result.

State A	State B	Overlap
Illinois	Indiana	0.6591306
Illinois	Wisconsin	0.6441679
Illinois	Michigan	0.6152271
Illinois	Ohio	0.5997812
Illinois	Utah	0.5995869
Illinois	Tennessee	0.5910998
Illinois	Nevada	0.5749631
Illinois	Missouri	0.5668692
Illinois	Kentucky	0.5504553
Illinois	North.Carolina	0.5426303

The Louvain algorithm works by treating $M^\top M$ as a weighted graph where each US state is a node, and edge weights are the cosine similarity scores computed above. It then iteratively reassigns states to communities by greedily maximizing modularity Q, which measures how much more densely connected states within the same community are compared to what you would expect by random chance, given their degree. In the first phase, each state starts in its own community; the algorithm repeatedly moves a state into whichever neighbor's community produces the largest gain in Q, until no single move improves it. In the second phase, each community is collapsed into a single "super-node" and the process repeats on this aggregated graph. This continues until Q can no longer be improved. The result is a partition of states into groups that share a high $M^\top M$ overlap, relative to the network as a whole.

Louvain is non-deterministic, so we run it 200 times with different random seeds to check whether the Illinois community assignment is stable or just a product of seed 42. Illinois and California end up in the same community in 0% of runs, and Illinois and New York in 0% of runs. Illinois is always in its own Midwest cluster regardless of seed. The partition is stable;

however, note that the modularity = 0.048 is low, meaning the community boundaries are real, but not sharp. In other words, states don't fall into tight, well-separated groups.

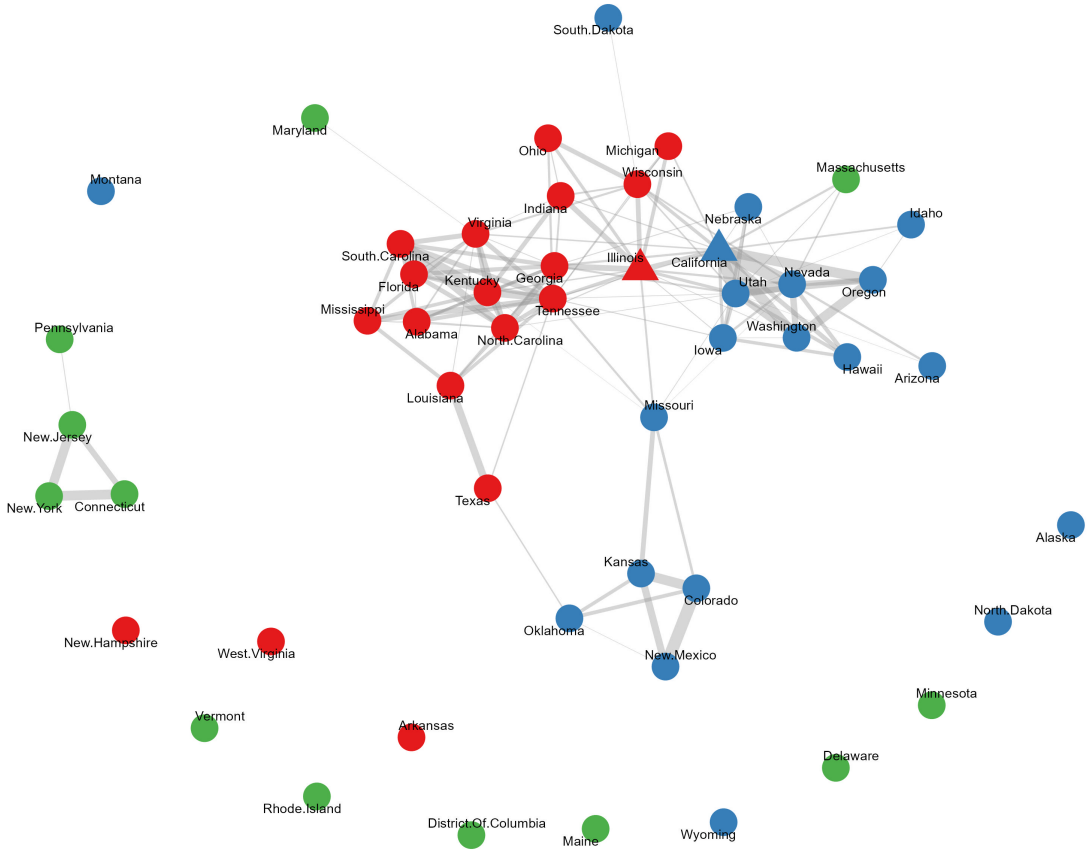


Figure 4: State Co-Exposure Network: Louvain Communities

Community	Number of states	States
1	19	Alabama, Arkansas, Florida, Georgia, Illinois, Indiana, Kentucky, Louisiana, Michigan, Mississippi, New Hampshire, North Carolina, Ohio, South Carolina, Tennessee, Texas, Virginia, West Virginia, Wisconsin
2	20	Alaska, Arizona, California, Colorado, Hawaii, Idaho, Iowa, Kansas, Missouri, Montana, Nebraska, Nevada, New Mexico, North Dakota, Oklahoma, Oregon, South Dakota, Utah, Washington, Wyoming
3	12	Connecticut, Delaware, District of Columbia, Maine, Maryland, Massachusetts, Minnesota, New Jersey, New York, Pennsylvania, Rhode Island, Vermont

4.4 MtM Co-Exposure Ranking for Illinois

The $M^\top M$ matrix gives us a direct prediction of which states should absorb the most redirected flow from an Illinois shock. The ranking shown below in Figure 5 is the empirical prediction of our propagation test. States near the top should show the largest reallocation toward them during the impasse.

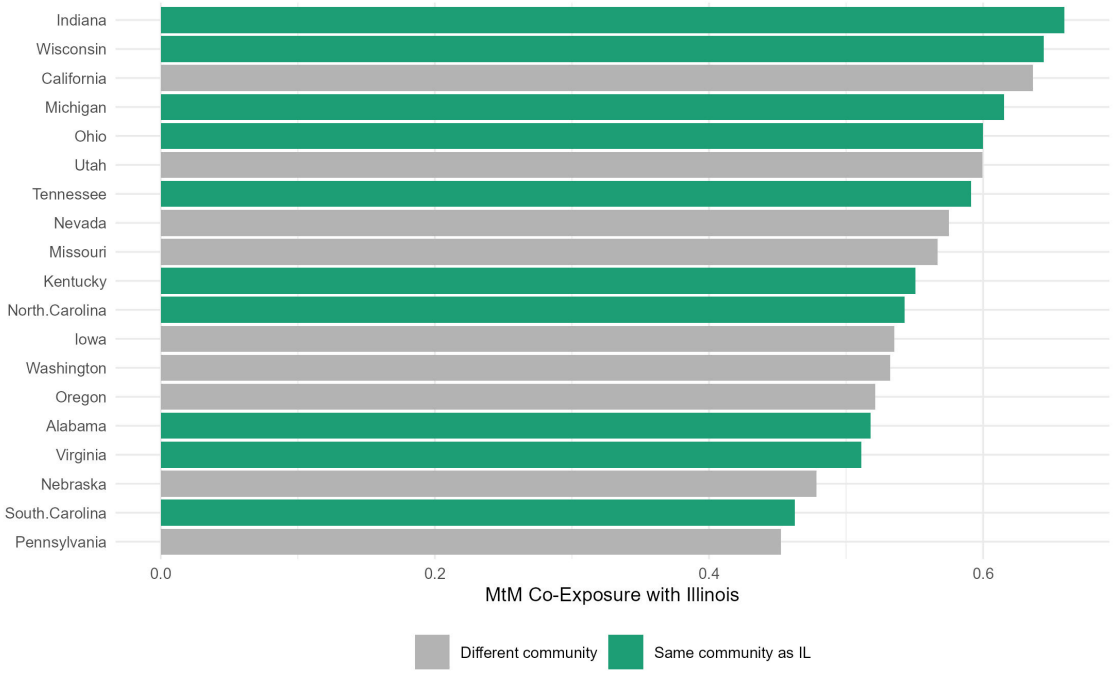


Figure 5: Predicted Cascade Targets: States Most Co-Exposed with Illinois

States in the same Louvain community as Illinois share the most sending municipalities and are the network's predicted absorption targets. States in different communities should show little re-routing even if they are large migration corridors in absolute terms.

5 Shock Detection: The Illinois Budget Impasse

The Illinois Budget Impasse ran from July 1, 2015 through July 10, 2017, 736 days in total. Without an operating budget, the state could not fund construction projects, pay contractors, or maintain social service contracts, the labor markets where Mexican migrants are concentrated.

Three features make this event attractive for causal identification:

Exogeneity. The impasse originated in a structural conflict between Governor Rauner and a Democratic supermajority legislature over pension reform and tax policy. Mexican migration decisions played no role in causing the crisis and were not targeted by any policy within it.

Specificity. No other US state experienced an equivalent fiscal event during 2015-2017. The top $M^\top M$ co-exposed states (California, Texas, Indiana) were in periods of fiscal stability or expansion, with no comparable labor-market disruption.

Timing. The impasse began July 1, 2015 and ended July 10, 2017. In annual MCAS data, this maps cleanly to treatment years 2015-2016, with 2017 as a partial-treatment year.

5.1 Exposure Variables

Not all Mexican municipalities are exposed equally to an Illinois shock. The key variable is each municipality's Illinois routing probability: of all migrants from municipality i, what fraction historically went to Illinois? We compute this from the pre-period weight matrix W^{pre} (averaged over 2010-2014 only), ensuring treatment assignment is not contaminated by migration responses that occurred during or after the impasse itself.

$$\text{Exposure}_i^{IL} = \frac{W_{i,IL}^{\text{pre}}}{\sum_j W_{ij}^{\text{pre}}}$$

This gives a continuous measure of pre-shock Illinois dependence. We complement it with the eigenvector centrality score computed in Section 2, which captures not just dependence on Illinois but the municipality's overall structural importance in the network.

Municipalities with zero Illinois routing probability ($N = 458$) have no predicted exposure to the shock and serve as the untreated comparison group in Tests 2 and 3.

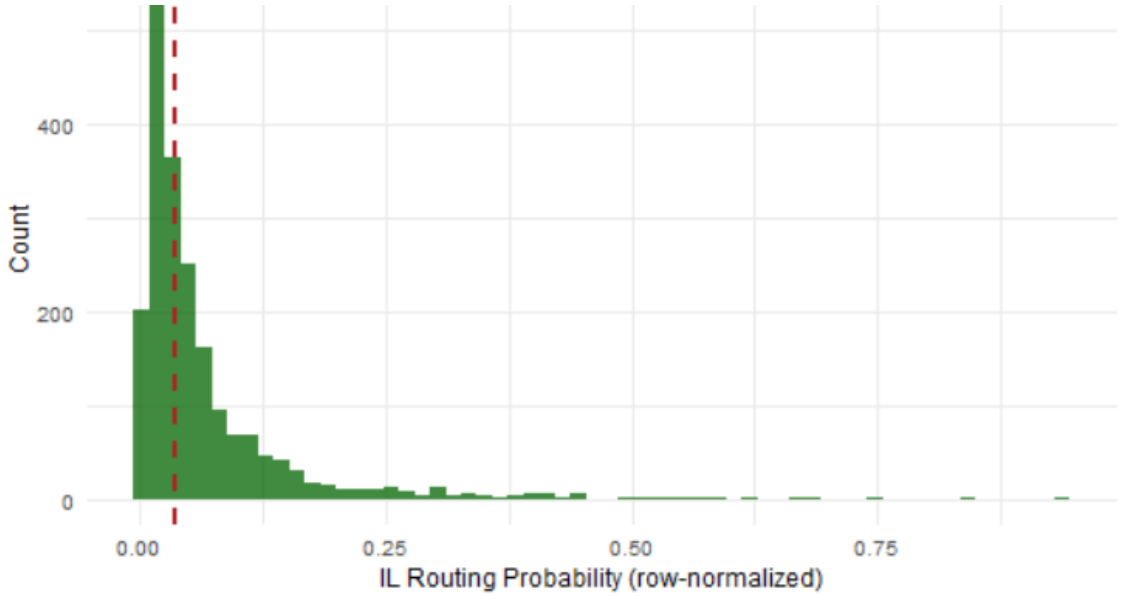


Figure 6: Distribution of Illinois Exposure Across MX Municipalities

6 Empirical Tests

The network gives us three concrete predictions to test. First, Illinois should show an anomalous drop in remittance outflows during the impasse relative to a counterfactual. Second, Mexican municipalities that routed more migrants through Illinois should have grown remittances more slowly than the rest during 2015-2017. Third, within those municipalities, the migration that did leave should have gone disproportionately toward states with high $M^\top M$ co-exposure with Illinois, the states that share the same sending communities.

We test all three with before-after comparisons: average outcomes during the impasse (2015-2017) minus the pre-period average (2012-2014). Each municipality or state is compared to its own baseline, so no parallel trends assumption is needed.

$$\Delta Y_i = \bar{Y}_i^{\text{impasse}} - \bar{Y}_i^{\text{pre}}$$

6.1 Test 1: Shock Identification

We use synthetic control method [Abadie et al., 2010] to test whether Illinois remittance outflows fell during the impasse. The idea here is to find the weighted combination of other US states whose pre-impasse remittance trajectory best matches Illinois's, then use that 'synthetic Illinois' as the counterfactual. The gap between actual and synthetic Illinois after July 2015 is the estimated impasse effect.

The synthetic control method here is meant to confirm to us that something unusual happened to Illinois remittances after July 2015, that a weighted average of other states would not have not otherwise predicted. In other words, it serves to validate our shock.

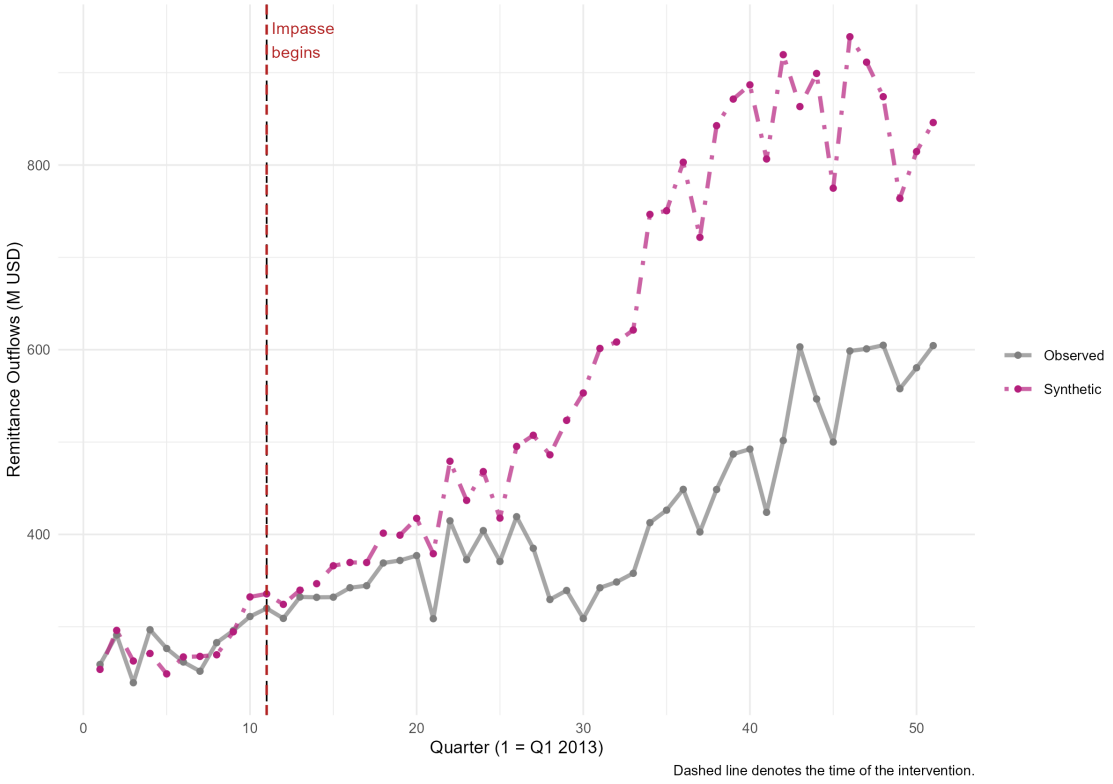


Figure 7: Synthetic Illinois v Actual Illinois Remittances, since Q1 2013

Figure 7 shows the synthetic control trajectory. For the pre-period, with only 10 quarters to match on, the optimization spreads weights nearly uniformly across donor states rather than finding a clean match. The post-period divergence is still visible: the synthetic climbs to roughly $900M by 2023 while actual Illinois stays around $600M.

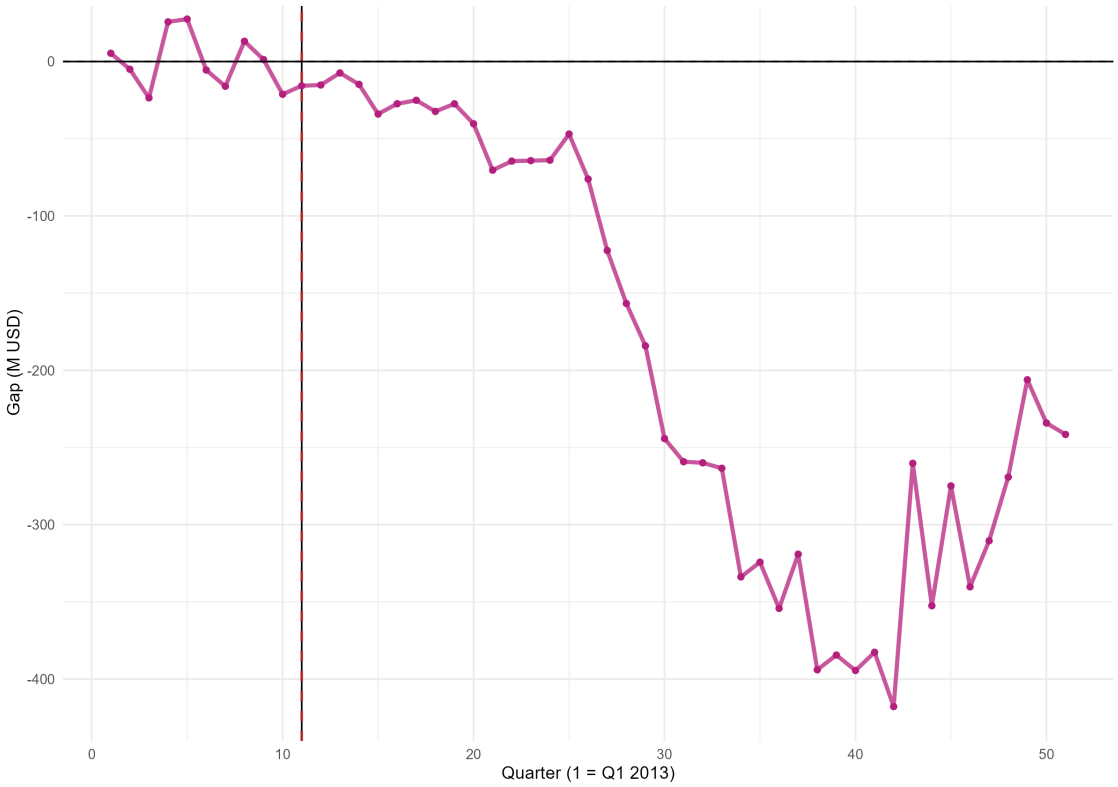


Figure 8: Illinois Impasse Effect: Actual Minus Synthetic Control

Figure 8 makes the gap explicit: before Q11 (Q3 2015) the 'actual-minus-synthetic' line sits near zero, confirming the pre-period match is acceptable. After treatment it turns and remains negative, reaching around -$400M at its deepest. The gap does not close after the impasse ended in Q3 2017 (Q19), which suggests either persistent labor-market effects or that the re-routing became sticky and never fully reversed back to Illinois. Note that high-MtM donor states absorbed the rerouted migrants from Illinois, so their DS2 outflows rose during the impasse. This inflates the synthetic upward and overstates the gap. Thus, the -$400M is not a conservative estimate; it is likely an overstatement of the true counterfactual divergence.

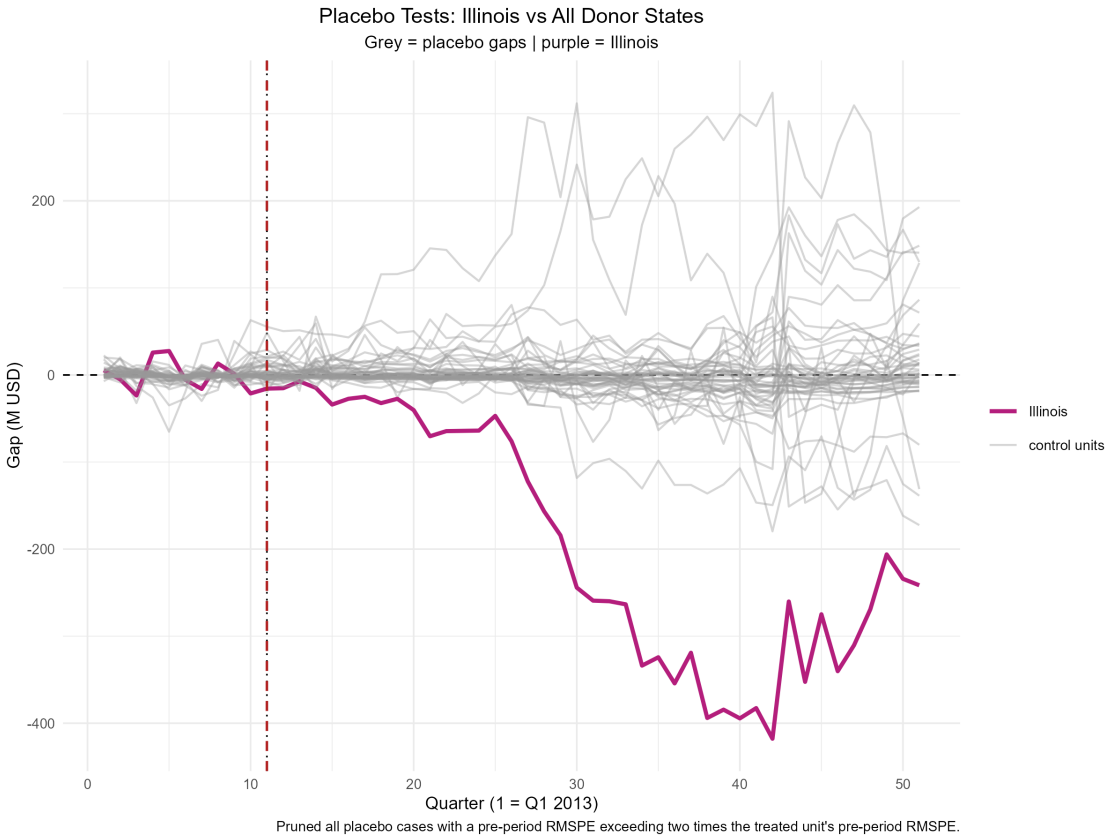


Figure 9: Placebo inference: same SCM run for every donor state. Illinois (purple) should sit outside the grey cloud for the result to be statistically meaningful. Poor-fit placebos (pre-RMSPE $>$ 2x Illinois pre-RMSPE) are pruned.

Figure 9 is the placebo test. Illinois (purple) sits well below every donor state throughout the post-period, ranking 3rd out of 52 on the post/pre MSPE ratio (permutation p = 0.058). The magnitude of the divergence is well outside what would be expected under the null of no treatment effect.

State	Weight
California	0.123
Florida	0.022
Colorado	0.019
Alabama	0.019
Arkansas	0.018
Arizona	0.018
Delaware	0.018
Connecticut	0.017
Georgia	0.017

The donor weight table shows the near-uniform distribution: California receives the highest weight at 12.3%, but the remaining states each receive under 2.2%, spread across structurally unrelated states like Alabama, Arkansas, and Delaware. This is a problem of the short pre-period: with 10 quarters and roughly 50 donors, the optimization is underdetermined and cannot isolate a clean synthetic match. California's prominence in the weights is worth noting: as the state with the highest $M^\top M$ co-exposure with Illinois, it was likely absorbing rerouted migrants during the impasse, which means its DS2 outflows were rising precisely because of the Illinois shock.

6.2 Test 2: Exposure Effect

US-Mexico remittances grew nationally during 2015-2017 (a secular trend unrelated to the impasse). This means the relevant question is not whether absolute remittances fell, but whether high-IL-exposure municipalities grew **less** than they otherwise would have. We regress each municipality's before-after log change on its pre-period Illinois routing probability:

$$\Delta \log \hat{R}_i = \alpha + \beta \cdot \text{IL_exposure}_i + \varepsilon_i$$

A negative β means higher IL exposure predicts slower remittance growth relative to the national trend. Zero would mean full re-routing with no income loss. Positive would mean IL-exposed municipalities actually benefited.

	Δ Log Remittances $\sim$ IL Exposure
(Intercept)	0.286^{***}
	(0.001)
il_exposure	-0.062^{***}
	(0.012)
Num.Obs.	2490
R^2	0.011
F	28.765

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

OLS

Outcome: log(avg R-hat 2015-17) − log(avg R-hat 2012-14)

The intercept of 0.286 gives us the national baseline: municipalities with no Illinois exposure grew remittances about 28.6% during the impasse period relative to pre-period. Remittances were rising across the board in 2015-2017. The slope of -0.062 is the cost of Illinois exposure on top of that. Higher Illinois routing probability predicts slower growth, around 6 percentage points per unit of exposure ($p < 0.001, N = 2,490$).

Figure 10 shows this result in practice. Every municipality in the sample is above the zero line, meaning they all grew in absolute terms. The impasse did not collapse remittances in these communities, it simply attenuated their growth relative to what they would have gotten otherwise. The R^2 of 0.011 is low, which is expected: the national trend accounts for most of the variance, and Illinois exposure explains only the margin around it.

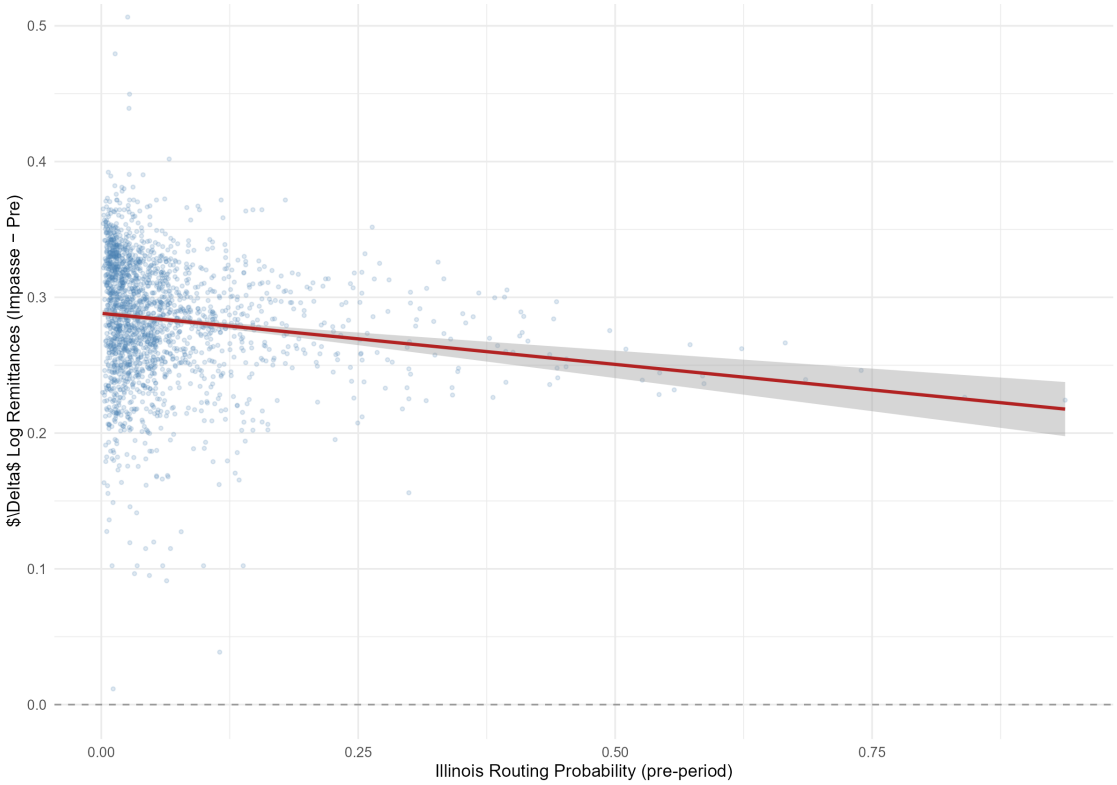


Figure 10: Municipality Exposure v. Remittance Change

6.3 Test 3: Cascade Ordering

If the impasse triggered re-routing rather than stopping, displaced migrants must have relocated somewhere else. The $M^\top M$ matrix provides for us a precise prediction: states with higher co-exposure to Illinois share more sending municipalities with it, so they are the most natural absorption targets.

We run this test at two levels of aggregation to show where the identifying variation lives: **state** and **municipality** level.

6.3.1 Test 3a: State-Level Aggregation

The natural first pass is a state-level before-after OLS: for each destination state j, compute the average change in migration share across high-IL-exposure municipalities and regress it on $M^\top M$ co-exposure with Illinois:

$$\Delta\bar{\text{Share}}_j = \alpha + \gamma \cdot [M^\top M]_{j,IL} + \varepsilon_j$$

Observations are weighted by pre-period average share x number of contributing municipalities to down-weight states with negligible absolute migrant flows.

	Δ Share$_j$ $\sim$ MtM$_j$ (state level)
(Intercept)	-0.002^{*}
	(0.001)
mtm_il	0.003
	(0.002)
Num.Obs.	47
R^2	0.037
F	1.735

Note: Weights = pre-period share $\times$ contributing municipalities.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The coefficient is positive but not significant ($p > 0.1$). With 47 observations and $M^\top M$ values in a narrow range, there is not enough power. But the bigger issue is that collapsing to state means throws away all the within-municipality variation. Municipalities vary a lot in total flow size, and averaging across them just lets the largest municipalities dominate rather than asking whether each one individually reallocated toward the right states.

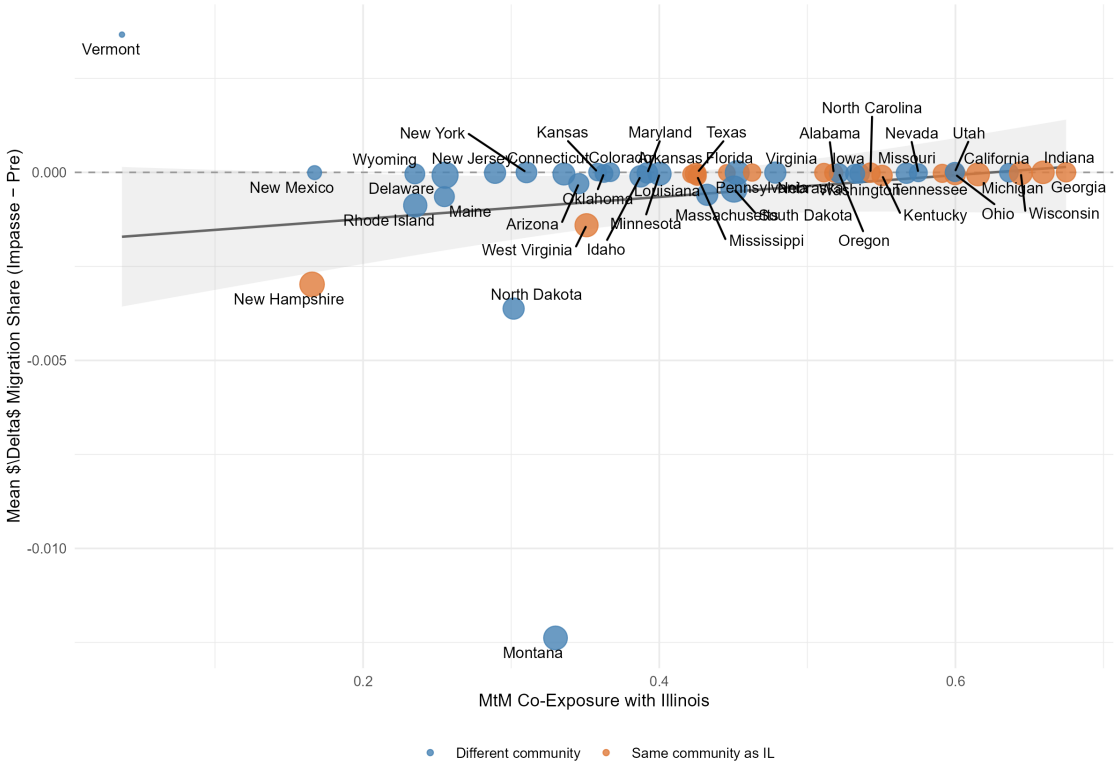


Figure 11: Test 3a: State-Level Cascade

6.3.2 Test 3b: Municipality × State Fixed Effects

The state-level test averages across thousands of municipalities before fitting the regression, losing the within-municipality reallocation signal. The correct unit of observation is **municipality x state**: for each treated municipality i and destination state j, did municipality i shift its migration toward states with higher $M^\top M$ co-exposure with Illinois?

Municipality fixed effects α_i absorb any total-flow shift at the municipality level (e.g. a municipality that sent fewer migrants everywhere equally). Two municipalities with very different total flows now contribute equally, so the coefficient is picking up whether the reallocation pattern within each municipality follows the $M^\top M$ ordering, not just whether the biggest municipalities happen to align with it.

$$\Delta\text{Share}_{ij} = \alpha_i + \gamma \cdot [M^\top M]_{j,IL} + \varepsilon_{ij}$$

	Δ Share$_{ij} \sim$ MtM$_j$ — Muni FE
mtm_il	0.000329***
	(0.000121)
Num.Obs.	21,903
R^2	0.029
FE: muni_id	X

Note: OLS with municipality fixed effects. Outcome: avg migration share to state j 2015-17 minus avg share 2012-14. Municipality FE absorbs total-flow shifts; coefficient tests reallocation within each municipality. SE clustered by municipality. Cosine-similarity MtM values range 0-1.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

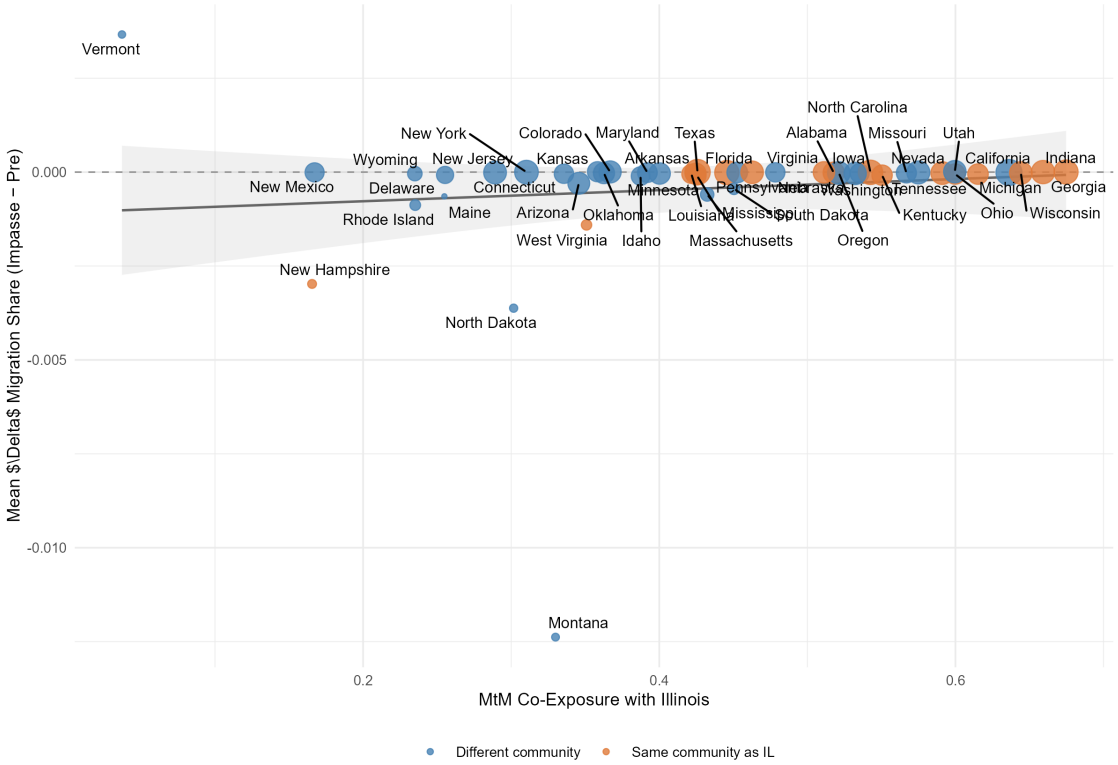


Figure 12: State-Level Means from the Municipality x State Sample

Within treated municipalities, states with higher $M^\top M$ co-exposure with Illinois absorbed a larger share of migration during the impasse ($\beta = 0.000329, p < 0.01, N = 21,903$). The coefficient is small in absolute terms because both migration shares and cosine similarity scores are fractions, but the direction is what the network predicted and it holds up once you control for total-flow shifts at the municipality level. The R^2 of 0.029 means $M^\top M$ accounts for about 3% of within-municipality reallocation variance. The rest is likely distance, existing labor-market ties, and personal networks, which are things the $M^\top M$ matrix does not capture.

The results from 3a and 3b point the same direction. The positive relationship between $M^\top M$ co-exposure and migration reallocation shows up at both levels, but only reaches significance once we stop aggregating. Collapsing 21,903 municipality x state observations into 47 state means wipes out the within-municipality variation that actually identifies the slope.

7 Discussion

In the first empirical test (Synthetic Control Method), Illinois remittance outflows fell well below the synthetic control from Q3 2015 onward, with a gap reaching roughly -$400M by 2017. Permutation p = 0.058 (rank 3 out of 52). This confirms the shock reduced Illinois remittance outflows. The -$400M should not be read as a precise causal estimate, just as a rough sense of the scale of the divergence.

In the second test (Exposure), municipalities more reliant on Illinois routing grew remittances about 6 percentage points slower than the national trend during the impasse (β = -0.062, p $<$ 0.001). The national baseline was strong growth (+28.6%), so these municipalities still grew, just less.

Finally, in the results within treated municipalities from the third test (Cascade), migration shifted toward states with higher $M^\top M$ co-exposure with Illinois (β = 0.000329, p $<$ 0.05, N = 21,903). The direction matches what the network predicts; the effect is small, however. $M^\top M$ explains about 3% of within-municipality reallocation variance.

8 Policy Implications

For Mexican sending communities, corridor disruptions cost sending municipalities real income even when migrants partially reroute. Municipalities with high Illinois routing probability grew remittances 6 percentage points slower during the impasse. For municipalities that depend heavily on US remittances, a labor-market shock in one destination state shows up quickly in household income.

For migrant-receiving states, a state that disrupts migrant labor conditions does not just retain fewer migrants; rather, it pushes them toward states that share the same sending communities. The $M^\top M$ ranking gives receiving states a way to see which of their neighbours' shocks they are most exposed to, before those shocks happen.

8.1 Limitations and Extensions

The paper has encountered a few caveats throughout the analysis. The first one is related to the methodological approach: the SCM pre-period is only ten quarters, so the gap estimate is approximate. MCAS only covers migrants who registered at a Mexican consulate, so if undocumented workers stopped registering during the impasse, Test 3's coefficient will be understated.

The other limitation found had to do with the R^2 in Test 3. It is 0.029, meaning $M^\top M$ explains about 3% of within-municipality reallocation. The effect is real but something else, probably distance, labor demand, and personal networks, accounts for most of where migrants actually went.

The analysis done in this paper could be extended by obtaining earlier DS2 datacto lengthen the SCM pre-period, linking MCAS to wage records to separate the re-routing channel from migrants who stayed in Illinois but earned less or running the same framework on other state-level shocks (e.g. Arizona's SB 1070) to see if $M^\top M$ co-exposure predicts cascade geography more generally.

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